

Ex.No.6	Use Sleuth Kit to analyze digital evidence
Date	

Description

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official [Sleuth Kit website](https://www.sleuthkit.org/) or https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device. Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command:

arduino

```
fsstat [image file] > filesystem_info.txt
```

- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:

arduino

```
mmls [image file] > partitions.txt
```

- This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command:

arduino

```
fls -r [image file] > file_list.txt
```

- This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:

css

```
icat [image file] [inode number] > [output file]
```

- Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command:

css

```
istat [image file] [inode number] > metadata_info.txt
```

- This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls:

css

```
fls -m / -r [image file] > body.txt
```

- Then create the timeline:

css

```
mactime -b body.txt > timeline.txt
```

- The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. Compile the Data:

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. Analyze and Document:

- Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

Output Screenshots:

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>dir mmls.exe
Volume in drive C is Windows-SSD
Volume Serial Number is 0A36-EC13

Directory of C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin

21-09-2026  19:39                75,264 mmls.exe
             1 File(s)                75,264 bytes
             0 Dir(s) 183,371,120,640 bytes free

C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>mmls.exe
Missing image name
Usage: mmls.exe [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-BrvV] [-aAmM] [-t vstype] image [images]
        -t vstype: The type of volume system (use '-t list' for list of supported types)
        -i imgtype: The format of the image file (use '-i list' for list supported types)
        -b dev_sector_size: The size (in bytes) of the device sectors
        -o imgoffset: Offset to the start of the volume that contains the partition system (in sectors)
        -B: print the rounded length in bytes
        -r: recurse and look for other partition tables in partitions (DOS Only)
        -v: verbose output
        -V: print the version
Unless any of these are specified, all volume types are shown
        -a: Show allocated volumes
        -A: Show unallocated volumes
        -m: Show metadata volumes
        -M: Hide metadata volumes

C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>
```

```
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5
```

CONTENT INFORMATION

```
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12    Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)       Size: 8-8      Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$LOGGED_UTILITIY_STREAM (256) Size: 0-65536   Flags: Non-resident
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>fls.exe -o 63 -r -d "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01"
r/- * 0: Documents and Settings/Default User/MPC7A4.tmp
r/- * 0: Documents and Settings/Default User/MPC7A5.tmp
r/- * 6366: Documents and Settings/Default User/MPC7A6.tmp
r/- * 6367: Documents and Settings/Default User/MPC7A7.tmp
r/- * 6368: Documents and Settings/Default User/MPC7A8.tmp
r/- * 6369: Documents and Settings/Default User/MPC7A9.tmp
r/- * 6370: Documents and Settings/Default User/MPC7AA.tmp
r/- * 6371: Documents and Settings/Default User/MPC7AB.tmp
r/- * 6372: Documents and Settings/Default User/MPC7AC.tmp
r/- * 0: Documents and Settings/Default User/MPC7B8.tmp
r/- * 0: Documents and Settings/Default User/MPC7B9.tmp
r/- * 6386: Documents and Settings/Default User/MPC7BA.tmp
r/- * 6387: Documents and Settings/Default User/MPC7BB.tmp
r/- * 6388: Documents and Settings/Default User/MPC7BC.tmp
r/- * 6389: Documents and Settings/Default User/MPC7BD.tmp
r/- * 6390: Documents and Settings/Default User/MPC7BE.tmp
r/- * 6391: Documents and Settings/Default User/MPC7BF.tmp
r/- * 6392: Documents and Settings/Default User/MPC7C0.tmp
r/- * 0: Documents and Settings/Default User/MPC7CC.tmp
r/- * 0: Documents and Settings/Default User/MPC7CE.tmp
r/- * 6408: Documents and Settings/Default User/MPC7CF.tmp
r/- * 6409: Documents and Settings/Default User/MPC7D0.tmp
r/- * 6410: Documents and Settings/Default User/MPC7D1.tmp
r/- * 6411: Documents and Settings/Default User/MPC7D2.tmp
r/- * 6412: Documents and Settings/Default User/MPC7D3.tmp
r/- * 6413: Documents and Settings/Default User/MPC7D4.tmp
r/- * 0: Documents and Settings/Default User/MPC7E0.tmp
r/- * 0: Documents and Settings/Default User/MPC7E2.tmp
r/- * 6428: Documents and Settings/Default User/MPC7E3.tmp
r/- * 6429: Documents and Settings/Default User/MPC7E4.tmp
r/- * 6430: Documents and Settings/Default User/MPC7E5.tmp
r/- * 6431: Documents and Settings/Default User/MPC7E6.tmp
r/- * 6432: Documents and Settings/Default User/MPC7E7.tmp
r/- * 6433: Documents and Settings/Default User/MPC7E8.tmp
r/- * 0: Documents and Settings/Default User/MPC7F4.tmp
r/- * 0: Documents and Settings/Default User/MPC7F6.tmp
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>icat.exe -o 63 "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01" 3726 > extracted_h323log.txt
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>dir extracted_h323log.txt
Volume in drive C is Windows-SSD
Volume Serial Number is 0A36-EC13
```

```
Directory of C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin
```

```
21-09-2026 20:00          0 extracted_h323log.txt
                1 File(s)          0 bytes
                0 Dir(s) 183,364,120,576 bytes free
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>notepad extracted_h323log.txt
```

```
C:\Users\dhani\Downloads\sleuthkit-4.15.0-win32\sleuthkit-4.15.0-win32\bin>istat.exe -o 63 "C:\Users\dhani\OneDrive\Attachments\Desktop\DellEvidence\4Dell Latitude CPI.E01" 3726-128-1
MFT Entry Header Values:
Entry: 3726          Sequence: 2
$LogFile Sequence Number: 11115366
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ( )
Created: 2004-08-19 22:47:33.514881600 (India Standard Time)
File Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
MFT Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
Accessed: 2004-08-19 22:47:33.514881600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: h323log.txt
Parent MFT Entry: 459 Sequence: 459
Allocated Size: 0 Actual Size: 0
Created: 2004-08-19 22:47:33.514881600 (India Standard Time)
File Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
MFT Modified: 2004-08-19 22:47:33.514881600 (India Standard Time)
Accessed: 2004-08-19 22:47:33.514881600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 88
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 148
Type: $DATA (128-1) Name: N/A Resident size: 0
```

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1.GitHub Activity & Submission Regularity	3	
2.Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The Sleuth Kit successfully analyzed the forensic disk image and identified file system and deleted file information. The metadata of the selected deleted file was examined using the istat command.